

Practical No. 05

DHCP Configuration on Cisco Packet Tracer

Easiest and complete revision notes, written for someone starting with zero networking knowledge

What This Practical Is About, in One Paragraph

In every earlier practical, you manually typed an IP address, subnet mask, and default gateway into each PC yourself, one by one. This practical teaches a way to stop doing that by hand. You set up a service called DHCP, either on a router or on a dedicated server, and once it is running, every PC on the network can simply say give me an address, and it receives one automatically, along with its subnet mask, default gateway, and DNS server, with zero manual typing. The whole purpose of this lab is to understand how that automatic handout actually works behind the scenes, and to configure it two different ways, once using a router, and once using a generic server device.

Key Terms Explained First, Before Anything Else

DHCP, Dynamic Host Configuration Protocol. A protocol whose entire job is to automatically hand out IP addresses and other network settings to devices, instead of a person typing them in manually on every single device. The word Dynamic is the key word here, it means the address is given out automatically and can change, as opposed to a fixed, manually typed address.

DHCP server. The device responsible for actually handing out the IP addresses. It can be a router, as done in the first part of this lab, or it can be a dedicated generic server device, as done in the second part of this lab. Either way, its job is the same, wait for a device to ask for an address, then hand one out.

DHCP client. Any device that receives its IP address automatically instead of being typed in manually. In this lab, every PC is a DHCP client once you switch its IP Configuration tab from Static to DHCP.

DHCP pool. A named, defined batch of IP addresses that the DHCP server is allowed to give out to clients. You give this pool a name yourself, for example MY_LAN, purely so you can identify it, the name has no technical effect on how it works.

Lease. The IP address a client is given is not owned by that client forever, it is only borrowed for a period of time, called a lease. Once the lease is close to running out, the client quietly asks to renew it, and normally keeps the same address without you noticing anything happen.

Excluded address. An address inside the DHCP pool's range that you tell the server never to hand out to anyone. This is done for addresses you want to keep fixed and manual, for example a printer, a server, or the router's own gateway address, since none of those should ever accidentally change.

Default gateway. The IP address a device sends its traffic to whenever the destination is outside its own network. Normally you would type this into every PC by hand, but with DHCP, the DHCP server itself tells every client what its default gateway should be, automatically.

DNS server, Domain Name System server. A server that translates human friendly website names into IP addresses. DHCP can also tell every client which DNS server address to use, again automatically, alongside the IP address and gateway.

Static IP address. An IP address that is manually typed in by a person and never changes on its own. This is the opposite approach to DHCP, and is still used for devices that must always be reachable at the exact same address, such as the DHCP server itself.

Generic server, in Packet Tracer. A device type in Packet Tracer meant to represent a real physical server. It can run different services, one of which is DHCP, turned on through its own Services tab, as an alternative to running DHCP directly on the router.

BOOTP. An older protocol that DHCP is considered an improvement of. BOOTP could hand out an IP address too, but in a more rigid, manual way, DHCP made this process automatic and flexible, which is why DHCP eventually replaced it.

ip helper-address. A special router command used when the DHCP server sits in a different network or broadcast domain than the clients asking for an address. Since a DHCP request is normally a broadcast, and broadcasts do not cross into another network by themselves, this command tells the router to forward that request onward to the real DHCP server elsewhere.

The Actual Workflow, Step by Step, and Why Each Step Happens This Way

This is the part most students memorize as commands without understanding why, so here is the reasoning behind every single

step, not just the step itself. When a PC's IP Configuration is set to DHCP, it does not yet have any IP address at all, so internally, four messages are exchanged between the client and the server before the PC finally has a usable address. This exchange is commonly remembered as DORA.

DISCOVER, OFFER, REQUEST, ACKNOWLEDGE

DISCOVER, the PC broadcasts a message to the entire network saying, I have no IP address, is any DHCP server out there.

OFFER, a DHCP server that hears this broadcast replies with an offer, here is an IP address you can use, along with a gateway and DNS server.

REQUEST, the PC replies back saying, yes, I accept that offered address, please give it to me formally.

ACKNOWLEDGE, the server confirms and finalizes the address, and from this point on the PC actually uses it.

Understanding DORA is the real reason behind why the DHCP server needs a pool, a gateway, and a DNS server defined ahead of time, all of that information has to already exist on the server before any PC ever sends its first Discover message, otherwise the server would have nothing correct to Offer.

Part One, Configuring DHCP Directly on a Router

The topology and the reasoning behind it

A router sits in the middle, connected to a switch, which connects to several PCs. The router's own interface facing the PCs, usually Fa0/0, is given a fixed IP address itself first, since the router needs a stable, known address of its own before it can act as the gateway and DHCP server for everyone else.

Step 1, Give the router's own interface a fixed address, this becomes the gateway

Command	What It Actually Does, and Why
enable	Moves into privileged mode, needed before deeper configuration is allowed.
config terminal	Enters global configuration mode, where interfaces and services are configured.
int fa0/0	Selects the router's interface facing the PCs, so the following commands apply to it specifically.
ip add 192.168.1.1 255.255.255.0	Gives this interface a fixed address. This exact address is what every PC will later automatically receive as its default gateway, so it must be set correctly before DHCP is even configured.
no shutdown	Activates the interface, router interfaces are off by default, this step is easy to forget and breaks everything else if skipped.
exit	Leaves interface configuration mode, back to global configuration mode.

Step 2, Build the actual DHCP pool on the router

Command	What It Actually Does, and Why
ip dhcp pool MY_LAN	Creates a new DHCP pool and gives it the name MY_LAN, purely a label you choose so you can recognize it, then enters DHCP pool configuration mode.
network 192.168.1.0 255.255.255.0	Defines the full range of addresses this pool is allowed to hand out, every usable address inside the 192.168.1.x network.
default-router 192.168.1.1	Tells the DHCP server which address to hand out to clients as their default gateway, this must match the router's own interface address from Step 1.
dns-server 192.168.1.10	Tells the DHCP server which address to hand out to clients as their DNS server, so their browsers know where to send name lookup requests.

Step 3, Exclude addresses that must never be handed out automatically

Command	What It Actually Does, and Why
ip dhcp excluded-address 192.168.1.1	Typed under global configuration mode, not inside the pool. It blocks this specific range from ever being offered to a DHCP client, protecting addresses that are fixed on purpose, such as the router's own gateway address and any statically addressed key devices like servers or printers.

192.168.1.10	
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Notice this exclude command sits in global configuration mode, one level above the DHCP pool itself, since it is a router wide rule, not something specific to just one pool.

Step 4, Switch every PC over to DHCP

1. Click on PC1, go to Desktop, then IP Configuration.
2. Switch the setting from Static to DHCP. The PC immediately runs through the DORA process in the background and should display an IP address, subnet mask, default gateway, and DNS server automatically, all matching what the router's pool was configured to hand out.
3. Repeat this for every remaining PC.

Step 5, Test it

From PC1, open Command Prompt and ping PC2's address. It should succeed, proving both PCs correctly received working addresses from the same DHCP pool and can reach each other normally.

Part Two, Configuring DHCP on a Generic Server Instead of a Router

The reasoning for this second method is simple, in real networks, DHCP is very often handled by a dedicated server rather than by the router itself, so this part of the lab shows the exact same idea, just configured through a different type of device, using menus instead of typed commands.

Step 1, Give the server itself a fixed IP address

Just like the router needed a fixed address before it could act as a DHCP server, the generic server also needs one. Set it to 192.168.1.2 with a subnet mask of 255.255.255.0, through its own Desktop or IP Configuration screen.

Step 2, Turn on and configure the DHCP service on the server

1. Click on the server, then open the Services tab.
2. Select DHCP from the list of services.
3. Fill in the DHCP settings, Pool name MY_LAN, Default Gateway 192.168.1.1, DNS Server 192.168.1.2, Start IP Address 192.168.1.0, Subnet Mask 255.255.255.0, Maximum Number of users 256.
4. Click Add, then Save, the new pool entry now appears in the service's list.
5. Make sure the DHCP service itself is switched to the On position, simply configuring the pool is not enough, the service also has to be actively running.

Here, the server itself is also acting as the DNS server, which is why the DNS Server field points back to the server's own address, 192.168.1.2, instead of a separate device.

Step 3, Switch every PC to DHCP again

Exactly as before, go to each PC's Desktop, then IP Configuration, and switch from Static to DHCP. All PCs should now automatically pick up an address from this server's pool instead of from a router.

An Important Addendum, Serving DHCP Across Different Networks

A DHCP Discover message is a broadcast, and broadcasts do not naturally travel across into a different network or broadcast domain. This becomes a real problem if your DHCP server sits in one network but your clients sit in a different one, behind a router. The fix for this situation is the **ip helper-address** command, configured on the router interface facing the clients, which tells the router, whenever you see a DHCP broadcast on this side, forward it directly across to the real DHCP server's address on the other side, instead of dropping it.

The Practice Exercise Mentioned in the Lab

The lab also includes a downloadable project file with a more advanced, real world style exercise, an ISP style router that must obtain its own public facing address using DHCP from the provider, while also acting as a DHCP server itself for its own internal PCs. The core ideas it tests are the exact same concepts explained above, just applied twice in the same topology, once

where the router is itself a DHCP client asking the ISP for an address, and once where the router is a DHCP server handing out addresses to its own internal PCs. It also introduces two very practical commands, **ipconfig /release**, which gives back a PC's current DHCP address, and **ipconfig /renew**, which asks for a new one from scratch, both useful for testing and troubleshooting real DHCP behaviour.

Troubleshooting, if a PC Does Not Get an Address

- Confirm no shutdown was typed on the router's interface facing the PCs, an inactive interface cannot deliver anything, including DHCP offers.
- Confirm the network command inside the DHCP pool matches the same network as the router's own interface address, a mismatch means the pool is handing out addresses from the wrong range entirely.
- Confirm the DHCP service on a generic server is actually switched On, a fully filled in pool that is still turned off will not respond to anything.
- Confirm the excluded address range is not covering the entire pool by mistake, if it excludes everything, there is nothing left to actually hand out.
- Give it a moment and try again, exactly like ARP, the first request after any change can occasionally need a second attempt.
- If clients are in a different network from the DHCP server, confirm ip helper-address is configured on the router interface closest to the clients.

Viva Questions for This Practical

Q1. What is DHCP and why is it useful?

Dynamic Host Configuration Protocol, a protocol that automatically hands out IP addresses and other network settings to devices, removing the need to manually type this information into every single device.

Q2. What is a DHCP pool?

A named, defined range of IP addresses that a DHCP server is allowed to give out to clients requesting an address.

Q3. What are the four steps of the DHCP process, and what do they stand for?

DORA, Discover, Offer, Request, Acknowledge. The client broadcasts a Discover message, a server replies with an Offer, the client sends a Request accepting it, and the server finalizes it with an Acknowledge.

Q4. Why does the router need a fixed IP address on its own interface before configuring DHCP?

Because that same address becomes the default gateway handed out to every client, it needs to be a known, unchanging address before the pool can reference it correctly.

Q5. What does the ip dhcp excluded-address command do, and where is it typed?

It reserves a range of addresses that the DHCP server must never hand out automatically, typically addresses that are fixed on purpose. It is typed in global configuration mode, not inside the DHCP pool itself.

Q6. Why would a network administrator want to exclude certain addresses from the DHCP pool?

To protect addresses assigned permanently to important devices such as servers, printers, or the gateway itself, so DHCP never accidentally assigns those same addresses to a random client.

Q7. What is the difference between configuring DHCP on a router versus on a generic server?

The underlying idea and the information handed out, pool range, gateway, DNS, are exactly the same, only the device doing the job differs, one uses CLI commands on a router, the other uses a graphical Services tab on a dedicated server device.

Q8. Why might a DHCP request fail to reach a DHCP server on a different network?

Because a DHCP Discover message is a broadcast, and broadcasts do not cross into a different network or broadcast domain on their own, without extra help from the router.

Q9. What command solves the problem of a DHCP server sitting in a different network than its clients?

ip helper-address, configured on the router interface facing the clients, so it forwards their DHCP broadcasts across to the real server.

Q10. What protocol did DHCP improve upon?

BOOTP, an older and more rigid protocol for handing out IP addresses.

Q11. What is a lease in the context of DHCP?

The period of time a client is allowed to keep and use an IP address before it must renew or give it up, since DHCP addresses are borrowed, not permanently owned.

Q12. What do the commands `ipconfig /release` and `ipconfig /renew` do?

`ipconfig /release` gives back the PC's currently held DHCP address, leaving it with none. `ipconfig /renew` then asks the DHCP server for a fresh address from scratch.

Q13. Does a DHCP server need to be given an IP address of its own?

Yes, whether it is a router interface or a generic server, the DHCP server itself always needs a fixed, static IP address, since it must always be reachable at a known address in order to answer requests.

Important Things to Remember for Future Labs and Exams

- Every DHCP client obtains four things automatically, an IP address, a subnet mask, a default gateway, and a DNS server, all in one exchange.
- The DORA sequence, Discover, Offer, Request, Acknowledge, is the actual mechanism behind every single DHCP assignment, this is one of the most commonly asked questions on this topic.
- The DHCP server itself must always have a fixed, static address, it is the one device on the network that cannot rely on DHCP for itself.
- Excluded addresses protect fixed, statically assigned devices from ever being accidentally handed out to someone else.
- DHCP broadcasts do not cross networks on their own, `ip helper-address` is the fix when the server and the clients are not on the same network.
- Configuring DHCP on a router and on a generic server achieve the exact same outcome, only the interface used to configure it is different, CLI commands versus a Services tab.
- Turning a pool's settings on is not enough by itself, on a generic server the DHCP service must also be explicitly switched On to actually start working.

Quick recall order: Fix the server or router's own address first, build the DHCP pool with gateway and DNS, exclude any reserved addresses, switch each PC to DHCP, then test with a ping.